

1. (2025 二中广雅周测)

【问题背景】在 $\triangle ABC$ 中, $AB=AC$, D 是 AC 上一点, E 是 BC 延长线上一点, 连接 DE , DB , 且 $\angle CDE=\angle ABD$.

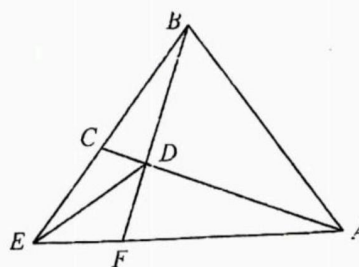
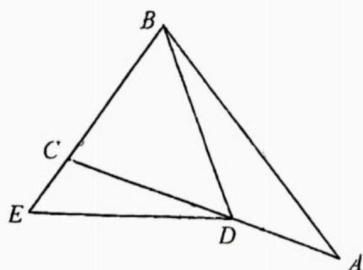
【问题探究】

(1) 如图 1, 直接写出 BD 与 DE 的数量关系是_____;

(2) 如图 1, 若 $BC=\sqrt{3}CE$, 求 $\frac{CD}{AD}$ 的值;

【拓展迁移】

(3) 如图 2, 延长 BD 交 AE 于点 F , 若 $AF=2EF$, 求 $\frac{AD}{AB}$ 的值.



【问题探究】

(1) 如图 1, 直接写出 BD 与 DE 的数量关系是 $BD=DE$;

(2) 如图 1, 若 $BC=\sqrt{3}CE$, 求 $\frac{CD}{AD}$ 的值: $\sqrt{3}-1$

【拓展迁移】

(3) 如图 2, 延长 BD 交 AE 于点 F , 若 $AF=2EF$, 求 $\frac{AD}{AB}$ 的值: $\frac{\sqrt{7}-1}{4}$

(2) 过 D 作 $DH \parallel AB$

设 $CE=1$, $BC=\sqrt{3}$

$\therefore BH=1$, $CH=\sqrt{3}-1$

$$\therefore \frac{CD}{AD} = \frac{CH}{BH} = \sqrt{3}-1$$

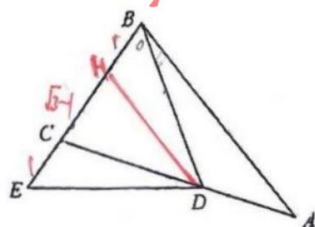


图 1

$$(3) \therefore \frac{DH}{AB} = \frac{CD}{AC}$$

$$\frac{a}{a+3} = \frac{3+a}{15+3a}$$

$$\therefore a = \frac{3\sqrt{7}-9}{4}$$

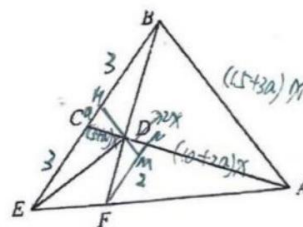


图 2

$$\therefore \frac{AD}{AB} = \frac{12+2a}{15+3a} = \frac{\sqrt{7}-1}{4}$$

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2. (武钢实验 5 月 8 号周测)

如图 1, 点 E 是 $\square ABCD$ 的边 AD 上一点, 连接 CE 并延长至点 F , 连接 FA 、 FD 、 BE , 使得 $\angle AFE=\angle BCF$, $DC=DF$.

(1) 求证: $\triangle ABE \cong \triangle FDA$;

(2) 求证: $\triangle AEF \sim \triangle BCE$;

(3) 如图 2, 连接 AC 交 BE 于点 G . 若 $AC \parallel DF$, 则 $\frac{AC}{DF} = \underline{\quad\quad}$.

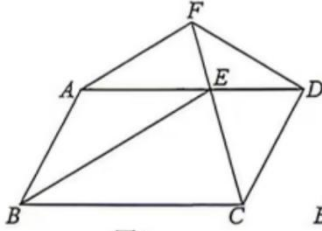


图1

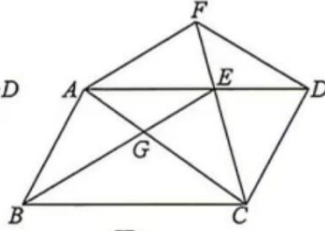


图2

23. (10分) 如图 1, 点 E 是 $\square ABCD$ 的边 AD 上一点, 连接 CE 并延长至点 F , 连接 FA , FD , BE , 使得 $\angle AFE = \angle BCF$, $DC = DF$.

(1) 求证: $\triangle ABE \cong \triangle FDA$;

(2) 求证: $\triangle AEF \sim \triangle BCE$;

(3) 如图 2, 连接 AC 交 BE 于点 G . 若 $AC \parallel DF$, 则 $\frac{AC}{DF} = \underline{\frac{1+\sqrt{5}}{2}}$.

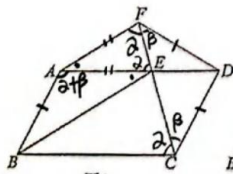


图1

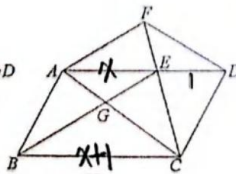


图2

(1) 证明: 在 $\square ABCD$ 中,
 $AB = CD$, $\angle DCB = \angle BAD$, $AD \parallel BC$
 $\therefore \angle AFE = \angle BCE = \alpha = \angle AFE$
 $\therefore \angle AFE = \angle AEF = \alpha$
 $\therefore AF = AE$

$\therefore DF = DC$,
 $\therefore \angle DFC = \angle DCF = \beta$
 $\therefore \angle AFD = \alpha + \beta = \angle BCD = \angle BAE$
 $\therefore AB = FD$, $\angle BAE = \angle DFA$, $AE = FA$
 $\therefore \triangle ABE \cong \triangle FDA (SAS)$

$$\frac{AE}{DE} = \frac{BC}{AE}$$

$$\therefore \frac{x}{1} = \frac{1+x}{x}$$

$$\therefore x^2 - x - 1 = 0$$

$$\therefore x = \frac{1 \pm \sqrt{5}}{2}$$

(2) 证明: $\therefore \triangle ABE \cong \triangle FDA$
 $\therefore BE = DA$
 $\therefore DA = BC$
 $\therefore BE = BC$
 $\therefore \angle BEC = \angle BCE = \alpha$
 $\therefore \angle AFE = \angle BEC$, $\angle AEF = \angle BCE$
 $\therefore \triangle AEF \sim \triangle BCE$
 (3) 解: $\therefore AC \parallel DF$
 $\therefore \triangle AEC \sim \triangle DEF$
 $\therefore \frac{AC}{DF} = \frac{EC}{EF} = \frac{AE}{DE}$
 $\therefore \triangle AEF \sim \triangle BCE$
 $\therefore \frac{EF}{EC} = \frac{AE}{BC}$
 $\therefore \frac{EC}{EF} = \frac{AE}{DE} = \frac{BC}{AE}$

3.(2026 年 71 本部五一作业) 已知正方形 $ABCD$, 点 F 是边 CD 上的动点 (不与点 C , D 重合), 点 E 在 BF 上.

(1) 如图 1, 连接 AE 并延长, 交边 BC 于点 G , 若 $AE \perp BF$, 求证: $AG = BF$;

(2) 如图 2. 当 $AE = AB$ 时, 连接 CE , 若 $CE \perp BF$, 求 $\tan \angle EBC$;

如图 3, 在矩形 $ABCD$ 中, 点 F 是边 CD 上的动点 (不与端点重合), 点 E 在 BF 上, 且 $AE = AB$, 连接 DE 并延长交 BC 于点 P , 若 $AB = 4$, $BC = 6$, 当点 F 为边 CD 的中点时, 求 $\frac{PE}{DE}$ 的值;

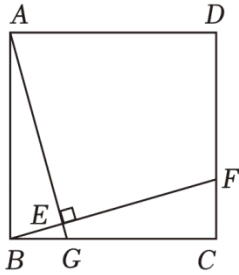


图1

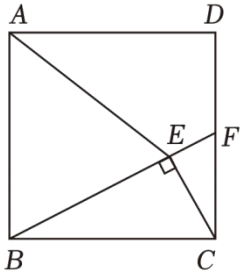


图2

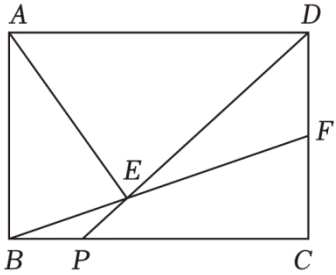


图3

第二问答案二分之一

(3)解：如图3， $AB = AE$ ，过点 A 作 $AH \perp BF$ ，垂足为 H ，延长 BF 交 AD 于点 Q ，则 $\angle AHB = 90^\circ$ ，

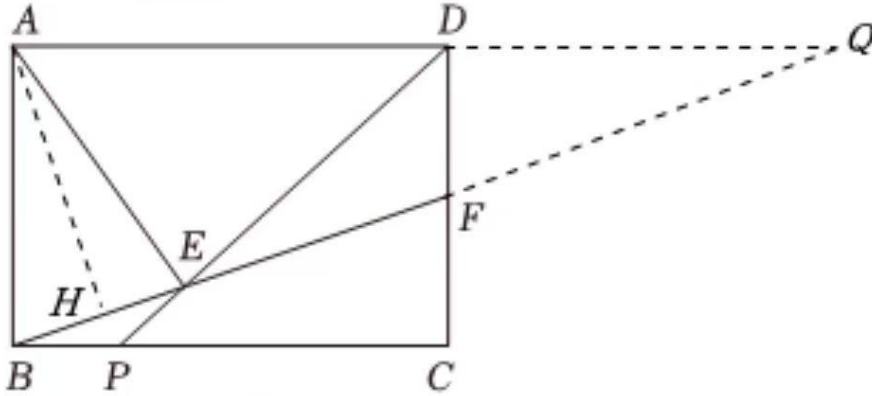


图3

$$\therefore EH = BH = \frac{1}{2}BE,$$

$\therefore$ 四边形 $ABCD$ 是矩形，

$$\therefore \angle ABC = \angle C = 90^\circ,$$

$$\therefore \angle ABH + \angle FBC = 90^\circ,$$

$$\therefore \angle ABH + \angle BAH = 90^\circ,$$

$$\therefore \angle FBC = \angle BAH,$$

$$\therefore \angle AHB = \angle C = 90^\circ,$$

$$\therefore \triangle AHB \sim \triangle BCF,$$

$$\therefore \frac{AH}{BC} = \frac{AB}{BF} = \frac{BH}{CF},$$

$\therefore AB = 4$, $BC = 6$, 当点 F 为边 CD 的中点，

$$\therefore CF = DF = \frac{1}{2}AB = \frac{1}{2}CD = 2,$$

在直角三角形 BCF 中，由勾股定理得： $BF =$

$$\sqrt{BC^2 + CF^2} = \sqrt{6^2 + 2^2} = 2\sqrt{10}.$$

$$\begin{aligned} \therefore AB &= 4, BC = 6, \text{ 当点 } F \text{ 为边 } CD \text{ 的中点,} \\ \therefore CF &= DF = \frac{1}{2}AB = \frac{1}{2}CD = 2, \end{aligned}$$

在直角三角形 BCF 中, 由勾股定理得: $BF = \sqrt{BC^2 + CF^2} = \sqrt{6^2 + 2^2} = 2\sqrt{10},$

$$\therefore \frac{AH}{6} = \frac{4}{2\sqrt{10}} = \frac{BH}{2},$$

解得: $BH = \frac{2\sqrt{10}}{5},$

$$\therefore BE = 2BH = \frac{4\sqrt{10}}{5},$$

$$\therefore AD \parallel BC,$$

$$\therefore \angle Q = \angle FBC,$$

$$\text{又} \therefore DF = CF, \angle QDF = \angle BCF = 90^\circ,$$

$$\therefore \triangle QDF \cong \triangle BCF (AAS),$$

$$\therefore QF = BF = 2\sqrt{10},$$

$$\begin{aligned} \therefore EQ &= BF + QF - BE = 2\sqrt{10} + 2\sqrt{10} - \\ &\frac{4\sqrt{10}}{5} = \frac{16\sqrt{10}}{5}, \end{aligned}$$

$$\therefore \angle Q = \angle EBP, \angle BEP = \angle QED,$$

$$\therefore \triangle BEP \sim \triangle QED,$$

$$\therefore \frac{PE}{DE} = \frac{BE}{EQ} = \frac{\frac{4\sqrt{10}}{5}}{\frac{16\sqrt{10}}{5}} = \frac{1}{4};$$

4. (经开外 5 月月考)

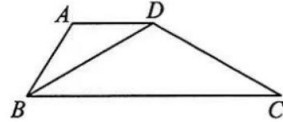
9. 如图①, 在梯形 $ABCD$ 中, $AD \parallel BC$, $AB = AD$, $\angle ABC = 2\angle BCD = 2\alpha$.

(1) 求证 $BD^2 = AD \cdot BC$.

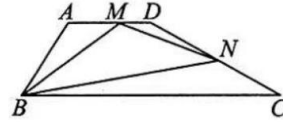
(2) 若点 M, N 分别在 AD, CD 上, 连接 BN , 且 $\angle BNC = \angle BMD$.

① 若 $\alpha = 30^\circ$ (如图②), 求证 $CN = \sqrt{3}MD$.

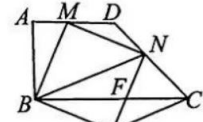
② 若 $\alpha = 45^\circ$, 以 BM 为边作正方形 $BMNE$, NE 交 BC 于点 F (如图③), 当 $AB = 3$, $MD = 2$ 时, 直接写出 $\triangle FEC$ 的面积是 _____.



图①



图②



图③

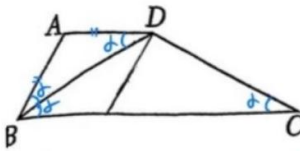
23. 如图, 梯形 $ABCD$ 中, $AD \parallel BC$, $AB = AD$, $\angle ABC = 2\angle BCD = 2\alpha$.

(1) 求证: $BD^2 = AD \cdot BC$;

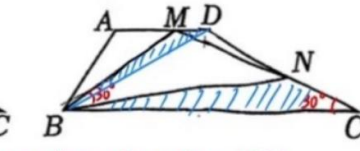
(2) 若点 M, N 分别在 AD, CD 上, 连 BN , 且 $\angle BNC = \angle BMD$.

① 若 $\alpha = 30^\circ$ (如图 2), 求证: $CN = \sqrt{3}MD$;

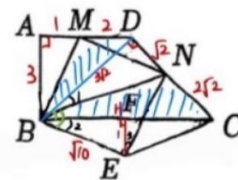
② 若 $\alpha = 45^\circ$, 以 BM 为边作正方形 $BMNE$, NE 交 BC 于点 F (如图 3). 当 $AB = 3$, $MD = 2$ 时, 直接写出 $\triangle FEC$ 的面积是 _____.



(1) $\triangle ABD \sim \triangle DBC$
 $\frac{AB}{BD} = \frac{BD}{BC}$ $\because AB = AD$
 $\therefore BD^2 = AD \cdot BC$



(2) $\angle BMD + \angle MBN + \angle CBN = 180^\circ$
 $\angle BNC + \angle C + \angle CBN = 180^\circ$
 $\Rightarrow \angle MBN = \angle C = 30^\circ = \angle DBC$
 $\Rightarrow \angle MBD = \angle CBN$
 $\Rightarrow \triangle BMD \sim \triangle BNC \Rightarrow \frac{DM}{CN} = \frac{BD}{BC} = \frac{1}{\sqrt{3}}$
 $\therefore CN = \sqrt{3}MD$



(3) $\alpha = 45^\circ \Rightarrow$ 由 (1) 得 $AB = AD = 3$ 且 $AB \perp AD$
 $BD = CD = 3\sqrt{2}$ 且 $BD \perp CD \Rightarrow BC = 6$
 $\therefore$ 可得 $\angle 1 = \angle 2 = 45^\circ \Rightarrow \tan \angle 1 = \tan \angle 2 = \tan 45^\circ = 1$
 $\therefore EH = 1, BH = 3, FH = \frac{1}{3} \Rightarrow CF = \frac{8}{3}$
 $\therefore S_{\triangle FEC} = \frac{1}{2} CF \cdot EH = \frac{4}{3}$

5. 大魏中考数学预测

23. (本题 10 分) 如图 1, 在矩形 $ABCD$ 中, 点 E 为 AB 的中点, 连接 CE , 过点 B 作 $BF \perp CE$ 于点 H , 交 AD 于点 F , 交 AC 于点 G .

(1) 求证: $AF \cdot AD = BE \cdot AB$;

(2) 若 $BC = nAB$, 求 $\frac{CG}{AG}$ 的值; (用含 n 的式子表示)

(3) 如图 2, 连接 AH , 在 (2) 的条件下, 直接写出 $\tan \angle EAH$ 的值. (用含 n 的式子表示)

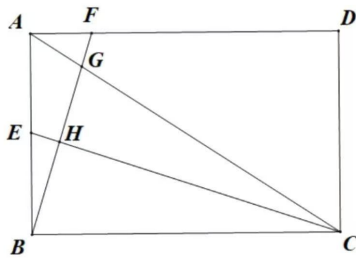


图 1

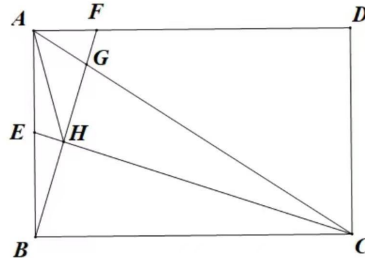
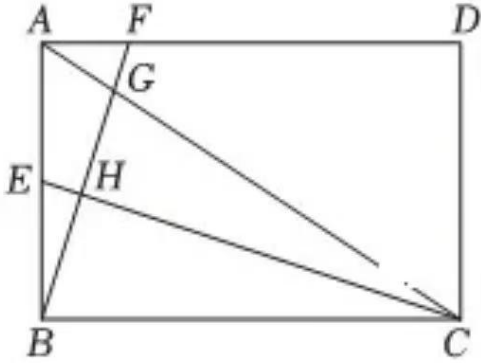


图 2

$$\therefore \triangle ABF \sim \triangle BCE,$$

$$\therefore \frac{AF}{AB} = \frac{BE}{BC} = \frac{BE}{AD},$$

$$\therefore AF \cdot AD = BE \cdot AB;$$



$$(2) \text{解: 由(1)得 } \angle AFB = \angle CBF, \quad \frac{AF}{AB} = \frac{BE}{BC},$$

$$\therefore AF = \frac{AB \cdot BE}{BC},$$

$$\because \angle AFB = \angle CBF, \quad \angle AGF = \angle CGB,$$

$$\therefore \triangle AGF \sim \triangle CGB,$$

$$\therefore \frac{CG}{AG} = \frac{BC}{AF} = \frac{BC^2}{AB \cdot BE},$$

$$\because BC = nAB, \text{ 点 } E \text{ 为 } AB \text{ 的中点,}$$

$$\therefore BE = \frac{1}{2}AB,$$

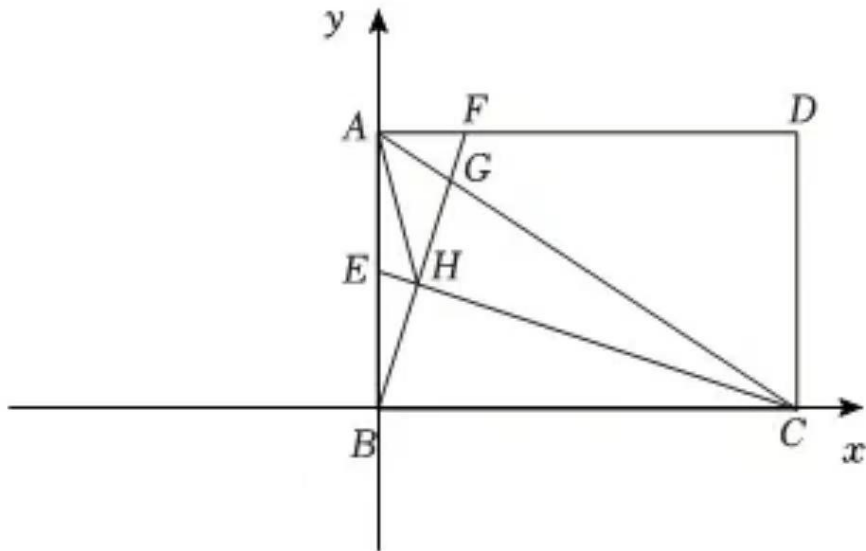
$$\therefore \frac{CG}{AG} = \frac{BC^2}{AB \cdot BE} = \frac{n^2 \cdot AB^2}{AB \cdot \frac{1}{2}AB} = 2n^2;$$

(3) 设 $AB = 1$, 则 $BC = n$, $BE = \frac{1}{2}$,

$$AF = \frac{AB \cdot BE}{BC} = \frac{1}{2n},$$

以 B 为原点, BA 为 y 轴, BC 为 x 轴, 则 $F(\frac{1}{2n}, 1)$

, $C(n, 0)$, $E(0, \frac{1}{2})$ 如图,



设 BF 所在直线的解析式为 $y = k_1x$ ($k_1 \neq 0$), 且经

过 $F(\frac{1}{2n}, 1)$, $B(0, 0)$,

$$\text{则 } 1 = \frac{1}{2n}k_1,$$

解得 $k_1 = 2n$,

$\therefore BF$ 所在直线的解析式为 $y = 2nx$,

$$\text{则 } 1 = \frac{1}{2n} k_1,$$

$$\text{解得 } k_1 = 2n,$$

$\therefore BF$ 所在直线的解析式为 $y = 2nx$,

设 CE 所在直线解析式为 $y = k_2x + b$, 且经过

$$E(0, \frac{1}{2}), C(n, 0),$$

$$\text{则 } \begin{cases} 0 = k_2n + b \\ \frac{1}{2} = b \end{cases},$$

$$\text{解得 } \begin{cases} k_2 = -\frac{1}{2n} \\ b = \frac{1}{2} \end{cases},$$

$$\therefore CE\text{所在直线解析式为 } y = -\frac{1}{2n}x + \frac{1}{2},$$

$$\text{当 } -\frac{1}{2n}x + \frac{1}{2} = 2nx \text{ 时, } x = \frac{n}{4n^2 + 1},$$

$$y = \frac{2n^2}{4n^2 + 1}, \text{ 即 } H(\frac{n}{4n^2 + 1}, \frac{2n^2}{4n^2 + 1}),$$

$$\tan \angle EAH = \frac{H_x}{1 - H_y} = \frac{\frac{n}{4n^2 + 1}}{1 - \frac{2n^2}{4n^2 + 1}} = \frac{n}{2n^2 + 1}.$$

6. 大魏中考数学预测

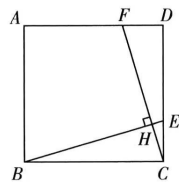
六、解答题(本大题共 12 分)

23. 在矩形 $ABCD$ 中, $BC = kAB$, 点 E 是 CD 边上不与端点 C, D 重合的动点, $CH \perp BE$.

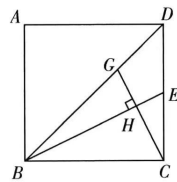
【课本再现】(1) 如图(1), 当 $k = 1$ 时, 延长 CH 交 AD 于点 F , 求证: $BE = CF$.

【类比迁移】(2) 如图(2), 在(1)的条件下, 连接 BD , 延长 CH 交对角线 BD 于点 G , 若点 E 是 CD 的中点, $EH = a$, 请分别求出 CH, BH 与 CG 的长(结果均用含有 a 的代数式表示).

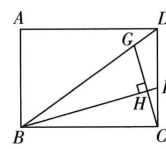
【拓展延伸】(3) 如图(3), 若 $DE = kCE$, 直接写出 $\frac{CG}{CH}$ 的值(结果用含有 k 的式子表示).



图(1)



图(2)



图(3)

$$\therefore BC = CD, \angle BCD = \angle D = 90^\circ,$$

$$\therefore \angle BCF + \angle DCF = 90^\circ.$$

$$\because CH \perp BE,$$

$$\therefore \angle BHC = 90^\circ,$$

$$\therefore \angle CBE + \angle BCF = 90^\circ,$$

$$\therefore \angle CBE = \angle DCF,$$

$$\therefore \triangle BCE \cong \triangle CDF (\text{ASA}),$$

(2 分)

$$\therefore BE = CF.$$

(3 分)

(2) 如图(1), 延长 CG 交 AD 于点 F ,

由(1)可知, $\triangle BCE \cong \triangle CDF$,

$$\therefore CE = DF, BE = CF.$$

$\because E$ 是 CD 的中点,

$$\therefore CD = 2CE,$$

$$\therefore BC = CD = 2CE = 2DF,$$

$$\therefore \tan \angle DCF = \tan \angle CBE = \frac{CE}{BC} = \frac{1}{2}.$$

$$\because EH = a, \therefore CH = 2a,$$

(4 分)

$$\therefore BH = 4a,$$

(5 分)

$$\therefore BE = BH + EH = 5a,$$

$$\therefore CF = BE = 5a.$$

(6 分)

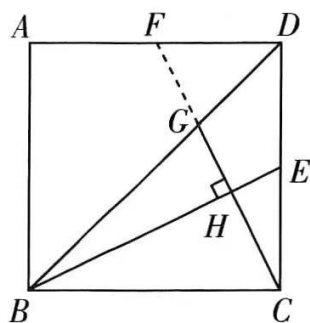
$$\because AD \parallel BC,$$

$$\therefore \triangle GFD \sim \triangle GCB,$$

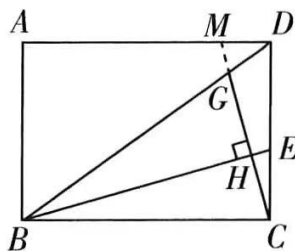
$$\therefore \frac{FD}{BC} = \frac{FG}{GC} = \frac{1}{2},$$

$$\therefore CG = \frac{2}{3}CF = \frac{10}{3}a.$$

(8 分)



图(1)



图(2)

$$(3) \frac{CG}{CH} = \frac{k^4 + 2k^3 + k^2 + 1}{k^3 + k^2 + 1}.$$

(12 分)

解法提示: 如图(2), 延长 CG 交 AD 于点 M .

同(1)可得, $\angle CBE = \angle DCM$.

又 $\because$ 在矩形 $ABCD$ 中, $\angle BCE = \angle CDM = 90^\circ$,

$$\therefore \triangle BCE \sim \triangle CDM,$$

$$\therefore \frac{BE}{CM} = \frac{CE}{DM} = \frac{BC}{CD} = k.$$

设 $DM = 1$, 则 $CE = k$,

$$\therefore DE = kCE,$$

$$\therefore DE = k^2,$$

$$\therefore CD = k^2 + k,$$

$$\therefore BC = kCD = k^3 + k^2.$$

$$\therefore \tan \angle CBE = \tan \angle DCM,$$

$$\therefore \frac{CH}{BH} = \frac{EH}{CH} = \frac{CE}{BC} = \frac{k}{k^3 + k^2} = \frac{1}{k^2 + k}.$$

设 $EH = b$,

$$\therefore CH = (k^2 + k)b, BH = (k^2 + k)^2 b,$$

$$\therefore BE = BH + EH = (k^2 + k)^2 b + b = (k^4 + 2k^3 + k^2 + 1)b,$$

$$\therefore CM = \frac{1}{k} BE = \frac{k^4 + 2k^3 + k^2 + 1}{k} b.$$

$\therefore$ 在矩形 $ABCD$ 中, $AD \parallel BC$,

$$\therefore \triangle BCG \sim \triangle DMG,$$

$$\therefore \frac{CG}{MG} = \frac{BC}{DM} = k^3 + k^2,$$

$$\therefore \frac{CG}{CM} = \frac{k^3 + k^2}{k^3 + k^2 + 1},$$

$$\therefore CG = \frac{k^3 + k^2}{k^3 + k^2 + 1} CM = \frac{k^3 + k^2}{k^3 + k^2 + 1} \cdot \frac{k^4 + 2k^3 + k^2 + 1}{k} b = \frac{(k^2 + k)(k^4 + 2k^3 + k^2 + 1)}{k^3 + k^2 + 1} b,$$

$$\therefore \frac{CG}{CH} = \frac{(k^2 + k)(k^4 + 2k^3 + k^2 + 1)b}{k^3 + k^2 + 1} \div [(k^2 + k)b] = \frac{k^4 + 2k^3 + k^2 + 1}{k^3 + k^2 + 1}.$$



名师讲方法

类比探究型问题的解题通法

类比探究型问题是共性条件与特殊条件相结合、由特殊情形到一般情形(或由简单情形到复杂情形)逐步深入、解题思路一脉相承的综合性题目. 解决类比探究型问题的一般方法:

(1) 根据题干, 结合分支条件解决第一问;

(2) 用解决上一问的方法类比解决下一问, 如果不能, 两问结合起来分析, 找出不能类比的原因和不变特征, 依据不变特征, 探索新的解题方法(如照搬字母, 照搬辅助线, 照搬全等或相似).

7. 大魏中考数学预测

七、解答题(满分 12 分)

25. 类比转化,从特殊到一般等思想方法,在数学学习和研究中经常用到,如下是一个案例,请补充完整. 原题:如图(1),在正方形 $ABCD$ 中,对角线 AC 、 BD 交于点 O ,点 E 是 BC 边上一点, AE 与 BD 交于 G ,过点 E 作 $EF \perp AE$ 交 AC 于点 F ,若 $\frac{BE}{CE} = 2$,求 $\frac{EF}{EG}$ 的值.

(1) 尝试探究

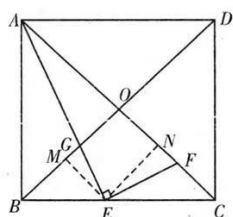
在图(1)中,过点 E 作 $EM \perp BD$ 于点 M ,作 $EN \perp AC$ 于点 N ,则 EM 和 EN 的数量关系是_____, $\frac{EF}{EG}$ 的值是_____.

(2) 类比延伸

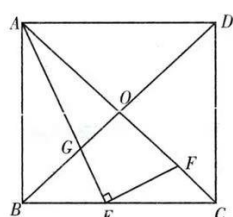
如图(2),在原题的条件下,若 $\frac{BE}{CE} = n (n > 0)$,则 $\frac{EF}{EG}$ 的值是_____ (用含 n 的代数式表示),试写出解答过程.

(3) 拓展迁移

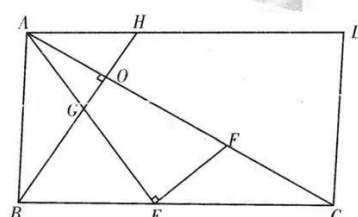
如图(3),在矩形 $ABCD$ 中,过点 B 作 $BH \perp AC$ 于点 O ,交 AD 于点 H ,点 E 是 BC 边上一点, AE 与 BH 相交于点 G ,过点 E 作 $EF \perp AE$ 交 AC 于点 F . 若 $\frac{BE}{CE} = a$, $\frac{BC}{AB} = b (a > 0, b > 0)$,求 $\frac{EF}{EG}$ 的值. (用含 a, b 的代数式表示)



图(1)



图(2)



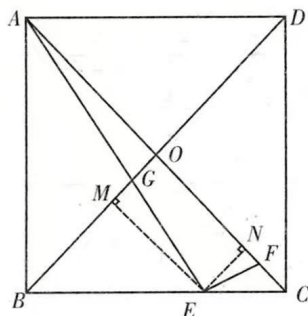
图(3)

$$\therefore \angle MEG = \angle NEF,$$

$$\therefore \triangle EMG \sim \triangle ENF,$$

$$\therefore \frac{EF}{EG} = \frac{NE}{ME} = \frac{1}{2}, \text{故答案为 } 2, \frac{1}{2}.$$

(1)



图(1)

(2) 解: $\frac{1}{n}$

过点 E 分别作 $EM \perp BD$ 于点 M, $EN \perp AC$ 于点 N.

$$\therefore \angle BMF = \angle CNE = 90^\circ$$

$\because$ 四边形 ABCD 是正方形, AC、BD 是对角线.

$$\angle OBC = \angle OCB = 45^\circ, \therefore \triangle BME \sim \triangle CNE$$

$$\therefore \frac{EM}{EN} = \frac{EB}{EC} = n, \therefore \angle MEG + \angle NEG = 90^\circ,$$

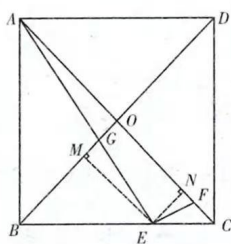
$$\therefore \angle NEF + \angle NEG = 90^\circ$$

$$\therefore \angle MEG = \angle FEN$$

$$\text{又 } \because \angle EMG = \angle ENF = 90^\circ, \therefore \triangle EMG \sim \triangle ENF$$

$$\therefore \frac{EM}{EN} = \frac{EG}{EF} = n$$

$$\therefore \frac{EF}{EG} = \frac{1}{n}$$



图(2)

(3)解: $\frac{1}{ab}$

分别作 $EM \perp BO$ 交 BO 于点 M , $EN \perp AC$ 于点 N .

$$\angle ENC = \angle BME = 90^\circ$$

又 $\because BH \perp AC$ 于点 O , 则 $EN \parallel BM$, $\therefore$

$$\angle NEC = \angle MBE$$

$$\therefore \triangle BME \sim \triangle ENC, \therefore \frac{BM}{EN} = \frac{BE}{EC} = a$$

又 $\because EN \perp AC$

$$\therefore \triangle CEN \sim \triangle CAB, \text{即} \frac{EN}{AB} = \frac{CN}{BC}$$

$$\therefore \frac{EN}{CN} = \frac{AB}{BC} = \frac{1}{b}$$

又 $\because \triangle BME \sim \triangle ENC$

$$\therefore \frac{BM}{ME} = \frac{EN}{CN} = \frac{1}{b} \text{ 即 } BM = \frac{ME}{b}$$

$$\therefore \frac{\frac{ME}{b}}{EN} = a \text{ 即 } \frac{ME}{EN} = ab,$$

$\because AE \perp EF, AC \perp BH$

$$\therefore \angle AOG = \angle AEF = 90^\circ$$

又 $\because \angle GAO = \angle FAE$,

$$\therefore \text{Rt} \triangle AGO \sim \text{Rt} \triangle AFE,$$

$$\therefore \angle AGO = \angle NFE$$

又 $\because \angle MGE = \angle AGO$,

$$\therefore \angle MGE = \angle NFE$$

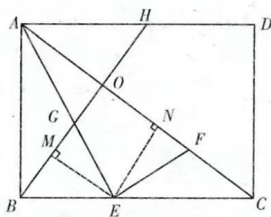
$\because EM \perp BO, EN \perp AC$

$$\therefore \angle EMG = \angle ENF = 90^\circ$$

$$\therefore \triangle EMG \sim \triangle ENF$$

$$\therefore \frac{EG}{EF} = \frac{EM}{EN} = ab$$

$$\therefore \frac{EF}{EG} = \frac{1}{ab}$$



图(3)